ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice b.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Drawings: award marks only for components that are drawn with the correct symbol and connected correctly. Accept other correct answers.

QUESTION 1

1.1 A capacitor is a component that stores electrical charge (energy) ✓ in an electric field between two plates separated by an insulator (dielectric) ✓. (2)

1.2 • Smoothing / filtering the output of a rectifier ✓ (2)
• Timing circuits (e.g. with a 555 timer) ✓
(Any TWO; also accept coupling / blocking DC, decoupling, motor starting.)

1.3 A relay is a small electromagnetic switch for low currents, used in control circuits. ✓✓ (4)
A contactor is a larger electromagnetic switch made to switch high currents, e.g. to motors, with arc suppression. ✓✓ (2 × 2)

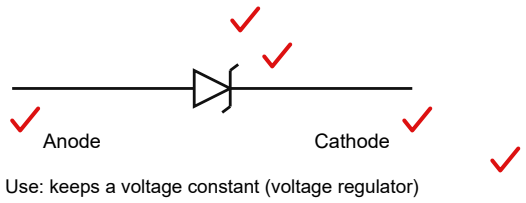
1.4.1 8 V ✓ (1)

1.4.2 2 × 8 = 16 V ✓ (1)

1.4.3 T = 5 ms ✓ (1)

1.4.4 $f = 1 \div T \checkmark = 1 \div 0{,}005 \checkmark = 200\ Hz \checkmark$ (3)

1.5



Use: keeps a voltage constant (voltage regulator)

(Diode triangle and bar 1, bent ends on the bar (Zener) 1, anode 1 and cathode 1 labelled; use, e.g. voltage regulator / voltage reference / over-voltage protection 2) (6)

[20]

QUESTION 2

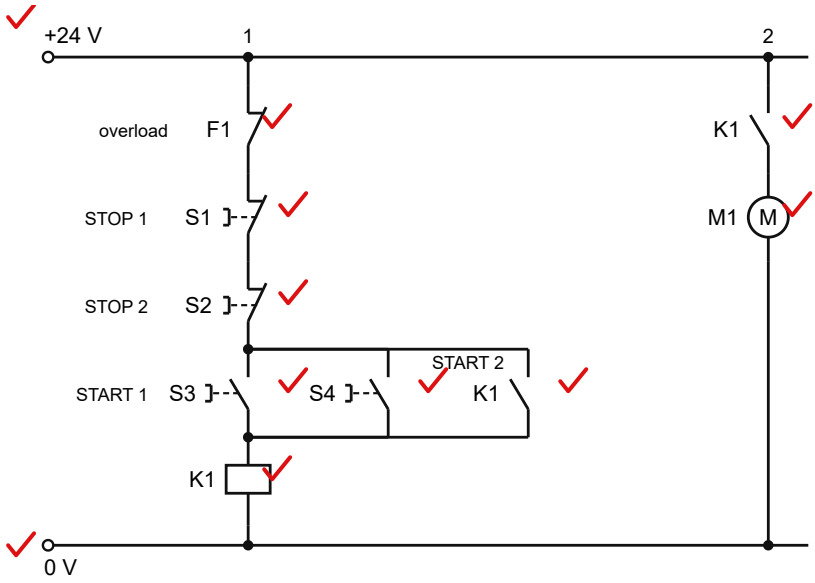
2.1 It protects the motor ✓ by opening the control circuit (stopping the motor) when the motor draws too much current for too long ✓. (2)

2.2 • Squirrel-cage induction motor ✓ (2)
• Slip-ring (wound-rotor) induction motor ✓
(Also accept synchronous motor, single-phase capacitor-start motor.)

2.3 A stepper motor turns in exact steps, so it can move to an exact position without a feedback sensor (open loop) ✓✓. (2)

2.4 It keeps the relay (motor) energised after the START button is released ✓, so the operator does not have to hold the button in ✓. (2)

2.5



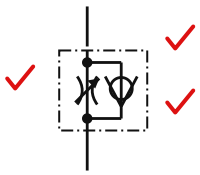
(Supply lines 1; overload contact F1 (NC) in series 1; STOP 1 and STOP 2 (NC) in SERIES 2; START 1 and START 2 (NO) in PARALLEL 2; holding contact K1 in parallel with the START buttons 2; relay coil K1 1; K1 contact switching the motor 2; neat, correct symbols 1)

(12)

[20]

QUESTION 3

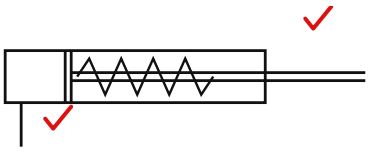
3.1.1



(Throttle (variable restriction) 1, check valve in parallel 1, dash-dot enclosure / correct flow direction 1)

(3)

3.1.2



(Cylinder with piston and rod and ONE port 1; spring on the rod side 1)

(2)

3.2 3.2.1 Pressure gauge ✓ (6)

3.2.2 Pressure relief valve ✓

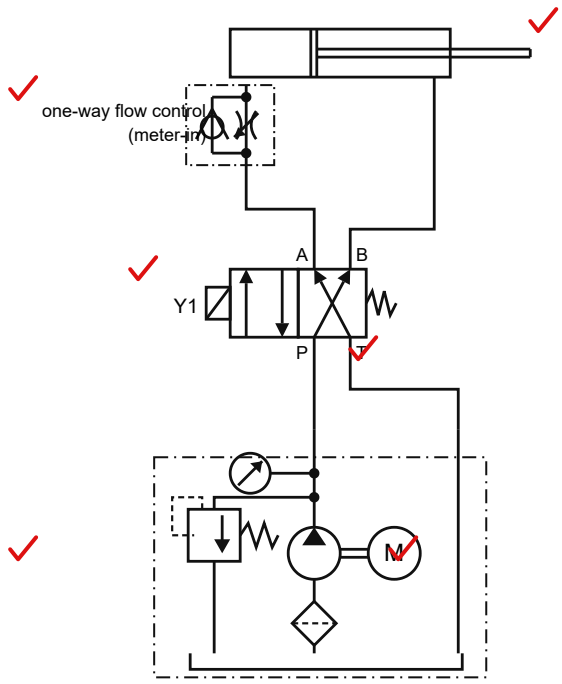
3.2.3 Reservoir (tank) ✓

3.2.4 Electric motor ✓

3.2.5 Hydraulic pump ✓

3.2.6 Suction filter ✓ (6 × 1)

3.3



(Power pack 2; 4/2 single-solenoid valve with spring return 2; double-acting cylinder correctly connected 1; one-way flow control in the line to the CAP end 2, with the check valve letting the oil out freely, so the oil flowing IN is throttled (meter-in) 2)

(9)

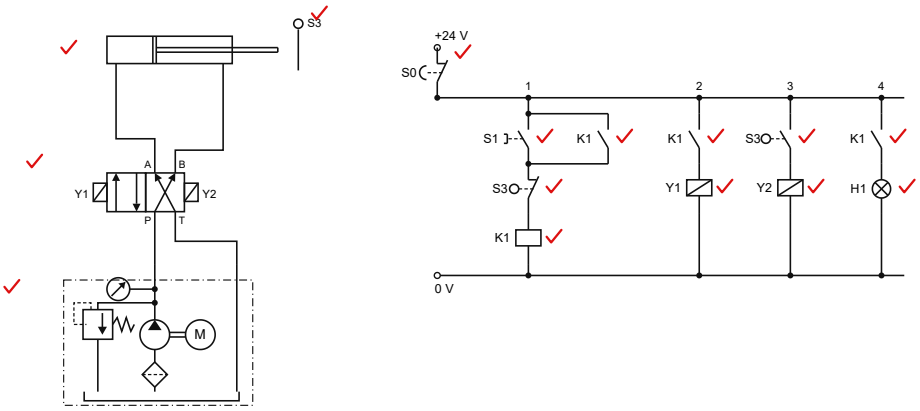
[20]

QUESTION 4

4.1 It switches an electrical contact when the oil pressure reaches a set value ✓, e.g. to stop the pump or signal the next step when the clamp pressure is reached ✓. (2)

4.2 • Mechanical limit switch (roller lever) ✓ (2)
• Inductive proximity sensor ✓
(Also accept magnetic reed switch on the cylinder, capacitive or optical sensor.)

4.3



(Working circuit: power pack 1; 4/2 double-solenoid valve Y1 / Y2 1; cylinder correctly connected 1; limit switch S3 at the end of the stroke 1. Control circuit: supply lines 1; E-stop (NC) in the supply 1; S1 (NO) 1 with K1 holding contact 1; S3 (NC) releasing K1 1; K1 coil 1; K1 contact switching Y1 1; S3 (NO) switching Y2 1; K1 contact switching lamp H1 1; Y1 and Y2 never on together 1; correct symbols 1; neat 1)

(16)

[20]

QUESTION 5

5.1 Maintenance done AFTER a fault or breakdown has happened ✓, to repair or replace the faulty part and return the machine to service ✓. (2)

5.2 Any FOUR ✓✓✓✓: oil level in the reservoir; no leaks at hoses and fittings; hoses not damaged or twisted; filters clean; guards in place; relief valve setting / gauge reads zero; emergency stop works. (4)

5.3 Any FOUR ✓✓✓✓: no supply / blown fuse; overload relay tripped; STOP or E-stop contact open (pressed or faulty); faulty START button; open-circuit relay coil; loose or broken wire; faulty relay contact; motor fault (burnt winding, worn brushes). (4)

5.4 (5)
- Isolate and lock out the supply and disconnect the motor leads ✓
- Set the tester to the correct test voltage (e.g. 500 V) ✓
- Test between each winding and the frame (earth) ✓ and between the windings ✓
- A high reading (e.g. more than 1 MΩ) means the insulation is good; a low reading means it is faulty ✓

5.5 Dirt in the oil wears pumps, valves and seals ✓, blocks small openings ✓ and causes valves to stick ✓. Any TWO ways ✓✓: change and clean filters regularly; keep the reservoir closed (breather filter); clean fittings before opening the system; use clean containers when topping up. (5)

[20]

TOTAL: 100

END OF MARKING GUIDELINE